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HW01

Content and Settings

Submissions (75)

Student Activity

Question Analysis

Homework Content



Question 1

10 points



Analog signal are signals that are:

- ☐ A. between 0 and 10 volts
- ☐ B. boolean operators (on/off)
- ☐ C. sampled signal
- ☒ D. Continuous functions of time

Correct answer

Question 2

10 points



Embedded systems are:

- ☐ A. systems that talk to the internet (internet of things)
- ☒ B. machines with micro controllers embedded into them

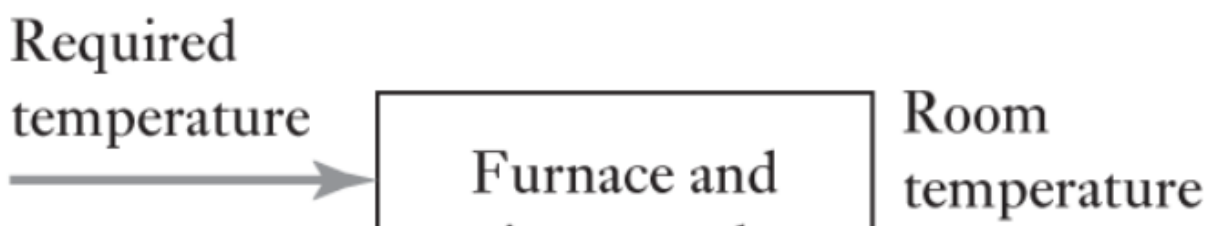
Correct answer

Question 3

10 points



The following picture is best defined as a:





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HW02

Content and Settings

Submissions (69)

Student Activity

Question Analysis

Homework Content

*All questions are randomly ordered for students*

Question 1

10 points



You develop a program that calculates taxes. What would be a proper variable name to express the adjusted gross income?

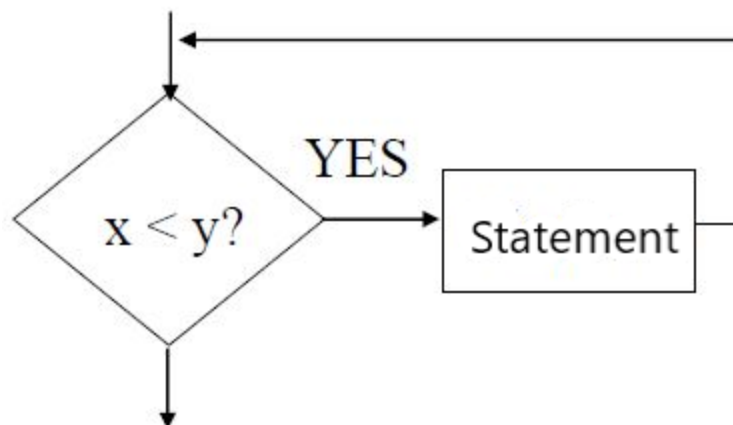
☒ A. adjustedGrossIncome

Correct answer

☐ B. x☐ C. adjusted income☐ D. int

Question 2

10 points





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HW03

Content and Settings

Submissions (73)

Student Activity

Question Analysis

Homework Content

*All questions are randomly ordered for students*

Question 1

10 points



The range of a transducer is the:

- ☒ A. Limit between which the input can vary. Correct answer
- ☐ B. The maximum value of the input minus the minimum value.
- ☐ C. The difference between the result of the measurement and the true value.
- ☐ D. The extent to which the value may be wrong.

Question 2

10 points



A thermocouple sensor has a time constant of 50 seconds in still air. When it is subject to a temperature change of 100°C, in 50 s its temperature will have changed by about:

- ☐ A. 31°C
- ☐ B. 50°C
- ☒ C. 63°C Correct answer
- ☐ D. 100°C

Question 3

10 points





Homework Content



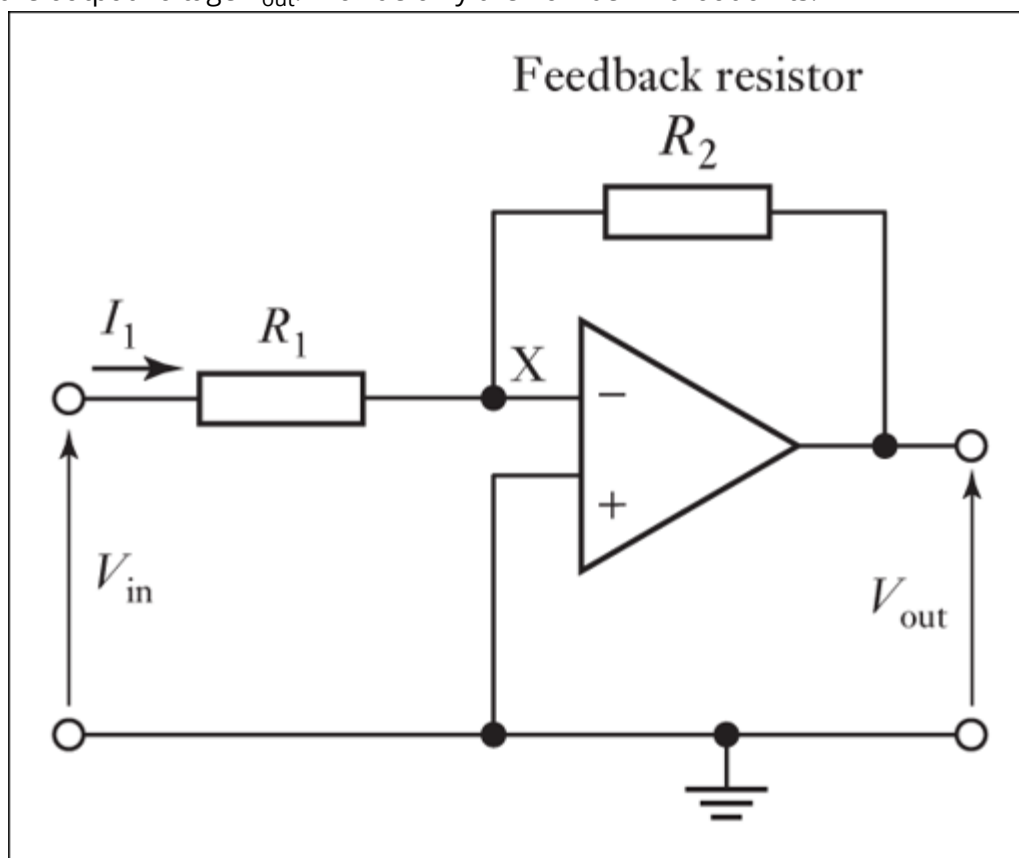
All questions are randomly ordered for students

Question 1

10 points



Consider the operational amplifier in the configuration below. If $V_{in}=5V$, $R_1=10\Omega$ and $R_2=50\Omega$, calculate the output voltage V_{out} . Provide only the number without units.



Question 2

10 points





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HW05

Content and Settings

Submissions (75)

Student Activity

Question Analysis

Homework Content

*All questions are randomly ordered for students*

Question 1

10 points



A multiplexer:

- ☐ A. Is a system that converts an analog signal, such as a sound picked up by a microphone or light entering a digital camera, into a digital signal.
- ☐ B. Is used for collecting the parity bits for validating the transfer of information.
- ☐ C. Is a system used for taking data from sensors and inputting that data into a computer for processing.
- ☒ D. Is a circuit that is able to have inputs of data from a number of sources and then, by selecting an input channel, give an output from just one of them. Correct answer

Question 2

10 points



The "quantization error" in a ADC comes from:

- ☐ A. The aliasing of the signal.
- ☒ B. The step like nature of digitizing the analog signal. Correct answer
- ☐ C. The quality of the ADC.
- ☐ D. It is not related to the ADC at all.



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HW06

Content and Settings

Submissions (73)

Student Activity

Question Analysis

Homework Content



Question 1

10 points



According to the lab policy (as posted on BB), who do you have to contact for any lab related questions\issues?

- ☐ A. Any TA of any session.
- ☐ B. My favorite TA.
- ☐ C. Just the professor.
- ☒ D. All the TAs of my lab section copying the professor in the email. Correct answer

Question 2

Full credit given

10 points



According to the lab policy (as posted on BB), what do you need to do if you cannot be physically present (with a valid reason) in one of the labs?

- ☐ A. Nothing, I can just show up next week.
- ☒ B. Contact all the TAs of my section and the section I wish to transfer to while Ccing the professor Correct answer
- ☐ C. Send a simple message that I will be absent.
- ☐ D. Attend a different section than the one I am registered.

Question 3

10 points





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HW07

Content and Settings

Submissions (65)

Student Activity

Question Analysis

Homework Content

*All questions are randomly ordered for students*

Question 1

10 points



The "V" flag tells the computer that:

☐ A. An interrupt was ignored.

☒ B. An overflow occurred.

Correct answer

☐ C. The result is a zero.

☐ D. A carry was generated.

Question 2

10 points



The accumulator register (A) is where :

☐ A. The status of the latest operation is indicated. This is done using flags with each flag being set or reset to indicate a specific status.

☐ B. A user enters values needed for computation.

☐ C. Information concerning the result of the latest process carried out in the arithmetic and logic unit is kept.

☒ D. Data for an input to the arithmetic and logic unit is temporarily stored.

Correct answer